Repository Supporting Tables

Reliability-First Evaluation for Spatio-Temporal Graph Forecasting

Table S1. Complete reference-degree controls

The ten cases below report changes in flat-pooled target MAE after zeroing the selected ten sensors across all input channels and history steps. Each case uses its own clean epoch-100 checkpoint reference. Uniform and degree-control values are means and sample SDs across 30 random draws, not retraining variability. The selected sensor set is fixed. The rank is one plus the number of degree-control draws with a smaller MAE change, among the selected value and 30 controls. Higher ranks indicate larger selected-set MAE changes relative to the degree controls. Rank 31 means a larger increase than all 30 controls, not better forecast accuracy. Ranks are descriptive, not significance tests.

Dataset	Model	Clean MAE	Selected Δ	Uniform Δ±SD	Degree Δ±SD	Rank
METR-LA	D2STGNN	2.879019	0.307359	0.302450 ± 0.073074	0.291343 ± 0.069949	22
PEMS-BAY	D2STGNN	1.514080	0.100838	0.124713 ± 0.015541	0.125062 ± 0.018997	5
PEMS04	D2STGNN	18.402961	4.105417	3.334588 ± 0.598201	3.178515 ± 0.393061	31
METR-LA	MTGNN	3.065592	0.670083	0.298490 ± 0.032341	0.338463 ± 0.056788	31
PEMS-BAY	MTGNN	1.602254	0.140866	0.115981 ± 0.014924	0.114733 ± 0.016701	27
PEMS04	MTGNN	19.098077	2.190102	2.147260 ± 0.328576	2.174683 ± 0.343091	17
METR-LA	MegaCRN	3.149723	0.176190	0.298734 ± 0.102925	0.318347 ± 0.114101	4
METR-LA	STGCN-Cheb	3.162829	0.071302	0.133999 ± 0.032916	0.126184 ± 0.029872	1
PEMS-BAY	STID	1.563499	0.146503	0.125829 ± 0.016737	0.114087 ± 0.014972	30
PEMS04	STID	18.427375	2.072549	3.471690 ± 0.543054	3.685005 ± 0.561158	1

Degree is measured in the supplied reference topology, not each architecture's learned or dynamic graph. Nine cases match selected degrees exactly. MTGNN on METR-LA uses degree 15 for one selected degree-18 slot because no unselected equal-degree sensor is available. The mean absolute degree mismatch is 0.3 across the ten slots.

These controls characterize checkpoint-specific target-error sensitivity and are interpreted separately from the sample-specific forecast-change diagnostics in the main manuscript. `Table S1.csv` identifies each source JSON, its SHA-256 and the rank field in the unchanged technical release `v1.4.4`.

Table S2. Calibration-boundary validity-mask check

This saved-array calculation keeps the original 3,415 calibration origins, ensemble members, fitted thresholds, interval definitions and target-validity mask unchanged. It excludes the first 11 evaluation origins, reducing the evaluation from 3,416 to 3,405 origins. No training or model inference is performed.

The first 11 evaluation origins contribute no valid target elements under the predefined mask because their target values are zero in the supplied arrays. The mask retains finite targets unequal to zero. These values match the corresponding supplied benchmark-array windows. Both evaluations therefore contain the same 7,118,436 valid target elements. Exclusion changes the origin count, not the evaluated target set.

	PICP (%)		MPIW	
Method	Original	Excluded	Original	Excluded
Dispersion-normalized, pooled	90.56704	90.56704	23.378166	23.378166
Dispersion-normalized, per horizon	90.56291	90.56291	23.180659	23.180659
Unscaled, pooled	90.14365	90.14365	13.789711	13.789711

MPIW uses the dataset's target units. Aggregate and all 12 horizon-wise PICP and MPIW values are unchanged for all three methods. The complete values are in `Table S2 all horizons.csv`. The corresponding JSON records the array hashes, metric comparisons and calibration quantiles. Recalculation reproduces the recorded aggregate and horizon-wise metrics.

This checks boundary handling for the retained METR-LA arrays and mask, not sensitivity to removing overlapping valid observations. It does not establish independence, exchangeability or conditional-coverage guarantees, or test the other datasets' boundary windows.